# MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

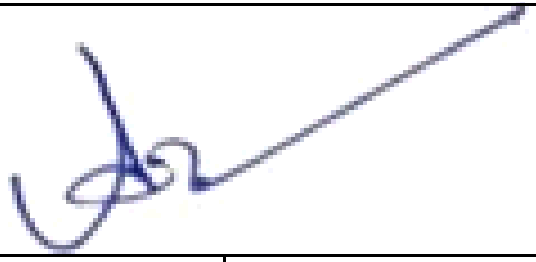
*(A constituent unit of MAHE, Manipal)*

## Master of Engineering - M.E in Computer Science and Engineering (Big Data Analytics)

## Course Plan

| | | |
|---|---|---|
| Course Name | : | Applied Machine Learning Lab |
| Course Code | : | AME 5052 |
| Academic Year | : | 2026-2027 |
| Semester | : | I |
| Name of the Course Coordinator | : | Dr. Arockiaraj |
| Name of the Program Coordinator | : | Dr. Richa Sharma |

| | |
|---|---|
|  | |
| Date: | 13-07-2026 |
| Signature of Program Coordinator | |
| with Date | |

| | |
|---|---|
|  | |
| Date: | 10-07-2026 |
| Signature of Course Coordinator | |
| with Date | |

## Table of Contents


| | |
|---|---|
| 1 Course Plan | 5 |
| 1.1 Primary Information | 5 |
| 1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping | 5 |
| 1.3 Assessment Plan | 6 |
| 1.4 Lesson Plan | 7 |
| 1.5 References | 9 |
| 1.6 Other Resources (Online, Text, Multimedia, etc.) | 9 |
| 1.7 Course Timetable | 10 |
| 1.8 Assessment Plan | 11 |
| 1.9 Assessment Details | 13 |
| 1.10 Course Articulation Matrix | 13 |

## Program Education Objectives (PEOs)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering - M.E in Computer Science and Engineering (Big Data Analytics), programare as follows

| PEO No. | Education Objective |
|---|---|
| PEO 1 | Develop in depth understanding of the key technologies in data engineering, data science and business analytics. |
| PEO 2 | Practice problem analysis and decision-making using machine learning techniques. |
| PEO 3 | Gain practical, hands-on experience with statistics, programming languages and big data tools through coursework and applied research experiences. |

## Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering - M.E in Computer Science and Engineering (Big Data Analytics), graduates will be able to:

| | |
|---|---|
| PO 1 | An ability to independently carry out research /investigation and development work to solve practical problems. |
| PO 2 | An ability to write and present a substantial technical report/document. |
| PO 3 | An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program. |
| PO 4 | An ability to develop and implement big data analysis strategies based on theoretical principles, ethical considerations, and detailed knowledge of the underlying data. |
| PO 5 | An ability to demonstrate knowledge of the underlying principles and evaluation methods for analyzing data for decision-making. |

# 1. Course Plan

## 1.1 Primary Information

| | | |
|---|---|---|
| Course Name | : | Applied Machine Learning Lab [AME 5052] |
| L-T-P-C | : | 3-0-0-3 |
| Contact Hours | : | 36 Hours |
| Pre-requisite | : | Python programming fundamentals |
| Core/ PE/OE | : | Core |

## 1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

| CO | At the end of this course, the student should be able to: | No. of Contact Hours | Program Outcomes (POs) | BL |
|---|---|---|---|---|
| CO1 | Apply different types of supervised machine learning algorithms to practical problems. | 12 | PO3 | 3 |
| CO2 | Apply different types of unsupervised machine learning algorithms to practical problems. | 12 | PO4 | 3 |
| CO3 | Analyse different types of machine learning paradigms. | 12 | PO1 | 4 |

## 1.3 Assessment Plan

| Components | Lab test | Flexible assessments | End semester exam (ESE) / Makeup exam |
|---|---|---|---|
| Duration | 90 minutes | 1 (15 days) | 180 minutes |
| Weightage | 0.3 | 0.2 | 0.5 |
| Typology of questions | Apply | Apply and analyse | Apply and analyse |
| Pattern | Answer all the questions. Maximum marks 30. | Answer all the questions. Maximum 20 marks. | Answer all the questions. Maximum marks 50. |
| Schedule | As per academic calendar. | 22-Oct-26 | As per academic calendar. |
| Topics covered | Introduction to Machine Learning, Decision Trees, Linear Models-Linear regression, Logistic regression. | Applyig and analysing supervised and unsupervised machine learning models. | Comprehensive examination covering the full syllabus. Students are expected to answer all questions. |

## 1.4 Lesson Plan

| L. No. | TOPICS | Course Outcome Addressed |
|---|---|---|
| L0 | Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books. | --- |
| L1 | Basics of Python for data handling: Introduction to Python libraries for machine learning (NumPy, Pandas, Matplotlib, Scikit-learn). | CO1 |
| L2 | Understanding datasets and data preprocessing techniques such as handling missing values, feature scaling, and encoding categorical variables. | CO1 |
| L3 | Implementation of decision tree classifier. Apply entropy, information gain and Gini impurity. Analyse the performance in terms of accuracy, precision, recall, F1-score. Apply cross-validation techniques. | CO1 |
| L4 | Analysis of overfitting and implementation of pruning techniques. Implementation of decision tree regressor. | CO1 |
| L5 | Implement linear regression model, apply different metrics (MSE, RMSE, R-sqaure). | CO2 |
| L6 | Apply logistic regression model-sigmoid function, decision boundary. Regularization methods (Ridge and Lasso). Analyse the performance in terms of accuracy, precision, recall, F1-score. Apply cross-validation techniques. | CO2 |
| LT | Lab Test | 0 |
| L7 | Apply K-NNN algorithms for the practical problems. Analyse performance - accuracy, precision, recall, F1-score, ROC curve and AUC. Cross-validation techniques. | CO2 |
| L8 | Implement maximum likelihood estimation (MLE) of parameters, Naive Bayes model for classification. Performance analysis and cross-validation techniques. | CO2 |
| L9 | Computing distances and similarities, implementation and analysis of K-means clustering and Hierarchical clustering, dendrogram construction and linkage methods | CO3 |
| L10 | Through coding, understand how ensemble methods in machine learning work, implementation and analysis of bagging techniques such as Random Forest and boosting algorithms. | CO3 |

| | | |
|---|---|---|
| L11 | Implementation of Gaussian Mixture Models (GMM) for clustering tasks. | CO3 |
| L12 | Implementation and visualization using Principal Component Analysis (PCA). Visualize, compare, and contrast different models for practical problems | CO3 |

## 1.5 References

1. G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, An Introduction to Statistical Learning: with Applications in Python, Springer Texts in Statistics. Cham, Switzerland: Springer, 2023. Online resource available at https://www.statlearning.com/
2. L. G. Serrano, Grokking Machine Learning, S. Thrun, Foreword, Manning Publications, 2021. Online resource from Manning Publications available at https://www.manning.com/books/grokking-machine-learning
3. J. VanderPlas, Python Data Science Handbook, Sebastopol, CA: O'Reilly Media, 2016.
4. A Course in Machine Learning, Hal Daumé III – Online resource available at http://ciml.info/
5. Module: Introduction to Machine Learning (https://www.intel.com/content/www/us/en/developer/tools/oneapi/training/academic-program/educators/intro-machine-learning-training-kit.html).
6. Learning path: Understand data science for machine learning (https://learn.microsoft.com/en-us/training/paths/understand-machine-learning/).

## 1.6 Other Resources (Online, Text, Multimedia, etc.)

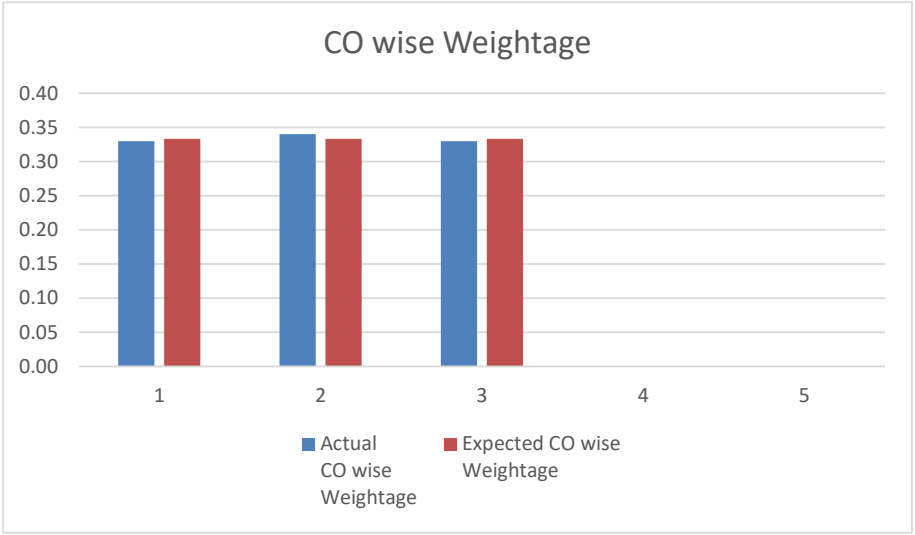
1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.

## 1.7 Course Timetable

| | Semester - I | | | | | Lab: | Data Science Lab | | |
|---|---|---|---|---|---|---|---|---|---|
| | 8-9 | 9-10 | 10-11 | 11-12 | 12-1 | 1-2 | 2-3 | 3-4 | 4-5 |
| Monday | | | | | | | | | |
| Tuesday | | | AML Lab | | | | | | |
| Wednesday | | | | | | | | | |
| Thursday | | | | | | | | | |
| Friday | | | | | | | | | |
| Saturday | | | | | | | | | |

## 1.8 Assessment Plan

| COs | | Marks & Weightage | | | | |
|---|---|---|---|---|---|---|
| **CO No.** | **CO Name** | **Lab test marks (Max. 30)** | **Assignment Marks (Max. 20)** | **ESE Marks (Max. 50)** | **Actual CO wise Weightage** | **Expected CO wise Weightage** |
| CO1 | Apply different types of supervised machine learning algorithms to practical problems. | 15 | 5 | 13 | **0.33** | **0.33** |
| CO2 | Apply different types of unsupervised machine learning algorithms to practical problems. | 0 | 10 | 24 | **0.34** | **0.33** |
| CO3 | Analyse different types of machine learning paradigms. | 15 | 5 | 13 | **0.33** | **0.33** |
| | | | | | | |
| | | | | | | |
| | **Marks (weightage)** | **0.3** | **0.2** | **0.5** | **1.00** | **1.00** |




Note:

- In-semester Assessment is considered as the Internal Assessment (IA) in this course for 50 marks, which includes performances in lab participation, assignment work, lab work, lab test, quizzes etc.
- ESE for this course is conducted for a maximum of 50.
- ESE marks for a maximum of 50 and IA marks for a maximum of 50 are added for a maximum of 100 marks to decide upon the grade in this course.

Weightage for CO1 = (Lab test marks for CO1 + Assignment marks for CO1 + ESE marks for CO1) /100
= (10 + 5+10)/100 = 0.25

## 1.9 Assessment Details

The assessment tools to be used for the Current Academic Year (CAY) are as follows:

| Sl. No. | Tools | Weightage | Frequency | Details of Measurement (Weightage/Rubrics/Duration, etc.) |
|---|---|---|---|---|
| 1 | Lab test | 0.3 | 1 | • Performance is measured using lab test attainment level.<br>• Reference: question paper and answer scheme.<br>• The lab test is assessed for a maximum of 30 marks. |
| 2 | Assignments | 0.2 | 1 | • Performance is measured using assignments attainment level.<br>• Assignment is evaluated for a maximum of 20 marks. |
| 3 | ESE | 0.5 | 1 | • Performance is measured using ESE attainment level.<br>• Reference: question paper and answer scheme.<br>• ESE is assessed for a maximum of 50 marks. |

## 1.10 Course Articulation Matrix

| CO | PO1 | PO2 | PO3 | PO4 | PO5 |
|---|---|---|---|---|---|
| CO1 | | | Y | | |
| CO2 | | | | Y | |
| CO3 | Y | | | | |
| | | | | | |
| | | | | | |
| Average Articulation Level | * | | * | * | |